Img16DatSer.exe
16-bit headerless DataSeries viewer (for Windows, Linux+WinE)

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1. The introduction

1.1 The scope & purpose

This utility is 16-bit <little-endian> header-less image viewer for Windows and (Ubuntu24,26) WinE.
Typical files' extensions are:
- *.bin - some cameras "o naturel";
- *.raw - ImageJ, AdobePhotoshop, Aldus Photostyler (note that these programs vary big|little-endianess); some cameras
- *.img, *.dat - some programs and cameras

= The main functionality of this utility:
- read 16bit little-endian pixels from image_file,
- hold the pixels in memory as 32bit signed ints,
- displays those as 8bit grayscale.

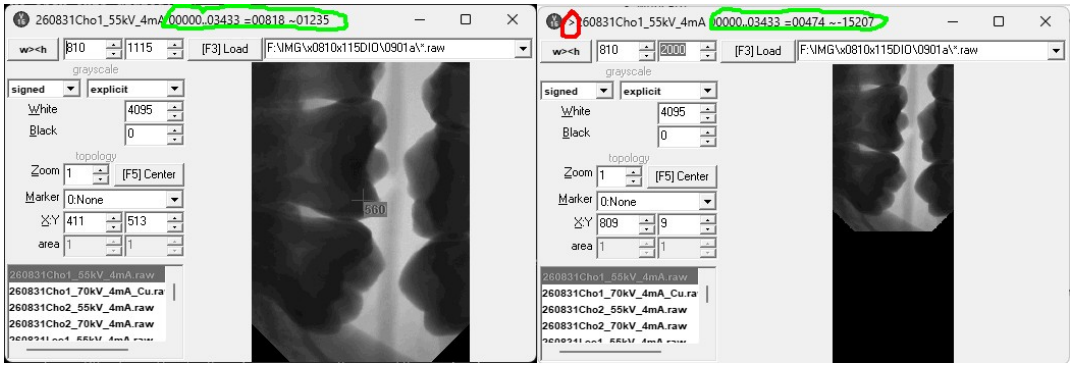
= Specific builds:
- "WindowsGDI" is default,
- "DirectX 2D" supports only "halftone bitblt-stretch"; this could become rational in ~ old Linux-PCs (when versatile WinE-GDI is too slow).

= Binary's name defines initial state of pixels' interpretation:
- default : - [signed],
- *U.exe, *u.exe: - [unsigned].

the input:
= section [View] in ini-profile
 - this_exe_directory/%IniProfileOfSpecificSDK% (if one exists),
 - this_exe_directory/this_exe_name.ini (otherwise);

= optional cmdline
 - "w h wildcard",
 - "wildcard",
 where wildcard composes as "directory/fname_wildcard" (e.g. ~"256 256 *.dat", ~"640 480 z:/img/1????.bin", ~"/home/user/img/*.raw").

window-title:
- <prefix> file_name min..max =median ~average
- '?'="no image", >="[w]*[h]>file_size",...



1.2 The development tools

= The projects are :
- EProg2008.vcproj : VC- build-toolchain (MS VS 2008)
- EProg.vcxproj : MS- build-toolchains (tested MS VS 2017,2002,2026)

= The character sets :
Assuming the WinE, build this project ~always as UNICODE. MBCS is possible for Windows (but ~has no purpose).

1.3 Usage of VSCode

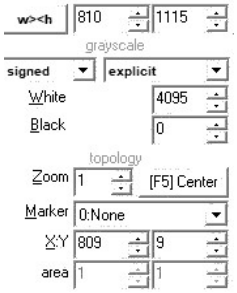
= File .vscode/c_cpp_properties.json

```
// this is rather a stub to have minimalistic code-highlights in VSCode
// and one-click build (when started under VS-developer-comman-prompt)
{  "version": 4,
   "configurations": [
      { "name": "Win64",
        "includePath": ["${workspaceFolder}/."],
        "defines": ["_WIN32"],
        "compilerPath": "cl.exe",
        "cStandard": "c17",
        "cppStandard": "c++17",
        "intelliSenseMode": "windows-msvc-x64",
```

```
        "configurationProvider": "ms-vscode.cmake-tools",
        "browse": {
          "path": ["${workspaceFolder}/**"],
          "limitSymbolsToIncludedHeaders": true
        }
      }
    ]
}
```

= There is no known tactic to make a robust CMakeLists.txt for a typical MFC-project.

2. The main View



2.1 View-controls

```
= [w><h]          :    - swaps the width,height

= [Load]          :    - browse & load image_files

= [combo box] (interprets pixels in image_files)
  - signed        :    - default -32768..32767
  - unsigned      :    - 0..65535 (when named as "View16U.exe")

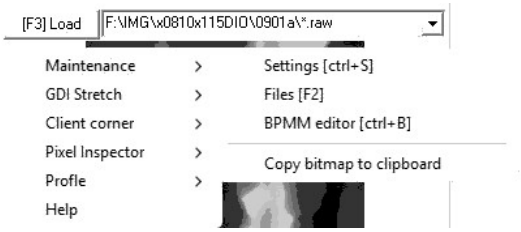
= [combo box] (gray_gamut)
  - explicit      :    - black=>RGB(0,0,0), white=>RGB(255,255,255)
  - THist         :    - cut 0.1% of pixels
  - inv THist     :    - cut 0.1% of pixels and invert

= [Zoom]          :    - magnifies frame's rectangle versus displayed one

= [combo Marker]:    - draws an accessory alongside image

= [edits X,Y]     :    - Marker's location

= [area]          :    - Marker's size
```



2.2 Pop-up menu

Appears by [F9], right mouse btn

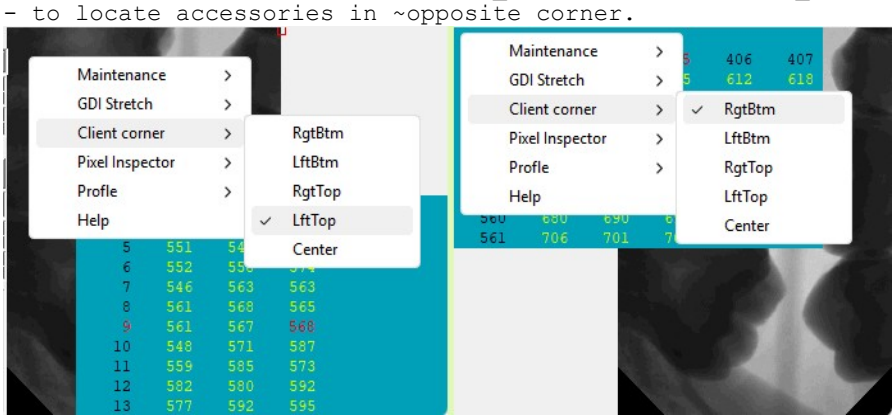
```
       Maintenance
= the dialogs (show particular help by [F1])
```

  - Settings   :   - UI-settings,
  - Files   :   - operate with items of files' list,
  - BPMM editor: - operate with BadPixels'Mask-overlays.

= the commands
  - Copy bitmap to clipboard,
  - Copy ROI text to clipboard,
  - Copy recent stats to clipboard.
  - [Execute]; the title might be 'Windows Explorer' or 'WinE Browser'.

GDI-stretch selection <when allowed>
- Fast   :   - StretchBlt(...COLORONCOLOR,)
- Halftone   :   - ...HALFTONE (the only option in DirectX-build),
- White|Black details: - ...WHITEONBLACK|BLACKONWHITE.

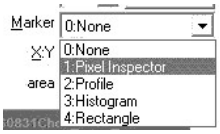
Client corner's selection
- to locate the image (when aspect_ratio reduces client_area),
- to locate accessories in ~opposite corner.



Pixel Inspector's selection
- 5x5 | 7x7 | etc.

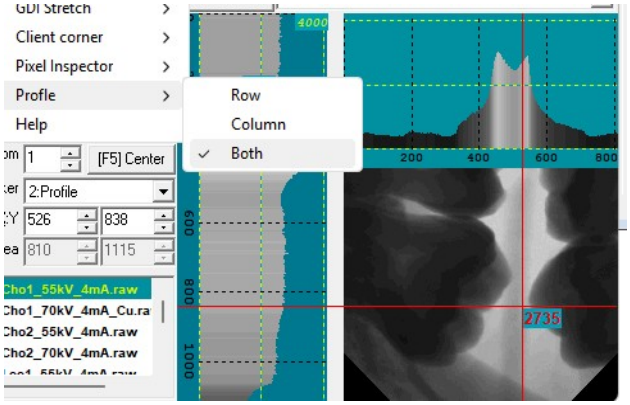
Profile's selection
- Row | Column | Both.

2.3 The markers

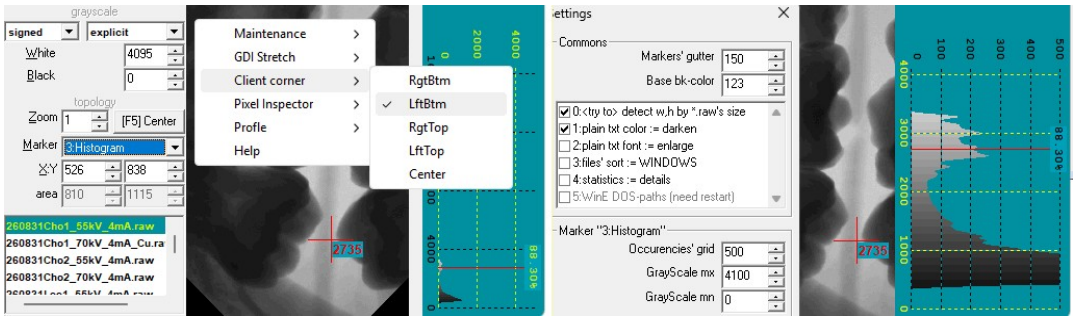


1:Pixel inspector
- ROI-center is view's spin-edits [X,Y]
- ROI-area is selection in pop-up menu

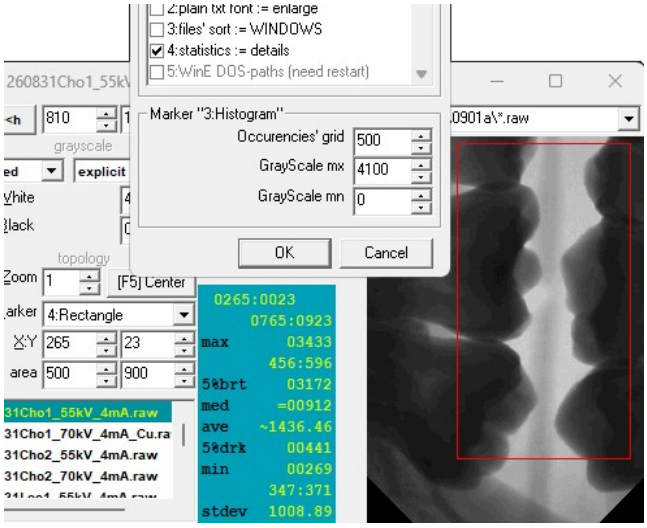
2:Profile
- the lines Row|Column|Both follow selection in the pop-up menu
- the grids follow the view



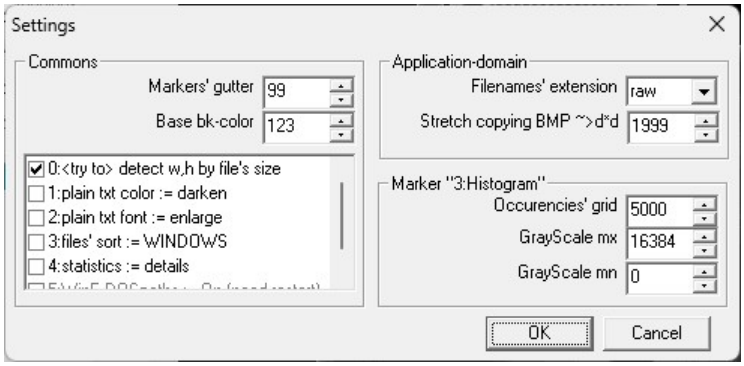
3:Histogram
- the grids follow [Settings]



4:Rectangle
- ROI-lft_top is view's [XY]
- ROI-area is view's [area
- presenting statistics follow [Settings]



3. Modal dialog [Settings]



3.1 settings "Commons"

```
= Markers' gutter   ~50..200
  <in pixels> width of markers 2:Profile, 3:Histogram.

= Base bk-color     ~50..200
  <in LSB> brightness of drawing accessories.

= Bit_flags
  - 0:detect w,h by file's size" - affects [Load];
the order of checks:
 a) the same FileSize in stored wildcards,
 b) if FS mod sqrt(FS)=0 then w:=h:=sqrt(FS),
 c) if FS mod w=0 then h:=FS div w,
 d) if FS mod h=0 then w:=FS div h.
  -  :"plain txt ... := ..."      - affect drawing accessories,
  - 3:"files' sort := WINDOWS"    - default is alphabet_order,
  - 4:"statiscics := details"     - affects markers 3:Histogram, 4:Rectangle,
  - 5:"WinE DOS-paths := On"      - show DOS-paths, force WinExplorer's stub
(affects under WinE).
```

3.2 settings "Application-domain"

```
= Filenames' extension:
  - *.raw ~ ImageJ, AdobePhotoShop, medical camers
  - *.bin ~ industrial cameras
  - etc

= Stretch copying BMP ~>d*d
  stretch CF_DIB for clipboard if exeeds this area (this mainly for WinE's
limitation). WinE might don't copy CF_DIB ~>4MB (i.e. set "d~<1999" for
grayscale 8-bit BMP in this application).
In Windows:
  - this adhoc limitation might be spractical;
  - one might assign large value (d~>9999) and copy large BMPs via clipboard
exactly.
```

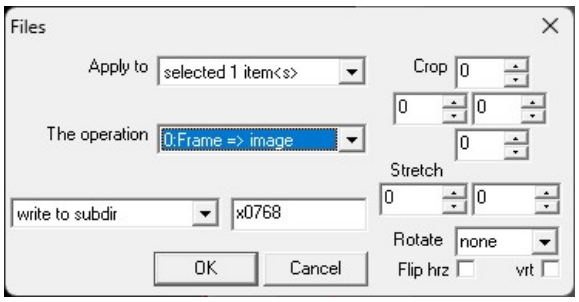
3.3 settings "marker 3:Histogram"

```
= Visible axes:
  - abscissa  - by "GrayScale_mn..mx",
  - ordinate  - by "0..Occurencies_grid".
```

4. Modal dialog [Files]

```
Terminology:
```

```
  - 'Frame' - inp image 'as is'
  - 'Image' - transformed 'Frame'.
```




```
      Apply to:
= current selection in files'list,
= whole files'list.

      The operations:
= 0:Frame => Image (in the following order):
  - Crop margins
  - Stretch to specified WxH
  - Rotate
  - Flip
  - Write ./SubDir/image_file or overwrite existing image_file

= 1:Write simple average frame '_ave'.

= 2:Frame => BMP (cut grayscale image_file<s> to 8-bit-gray *.BMP).

= 3:Delete file<s>.

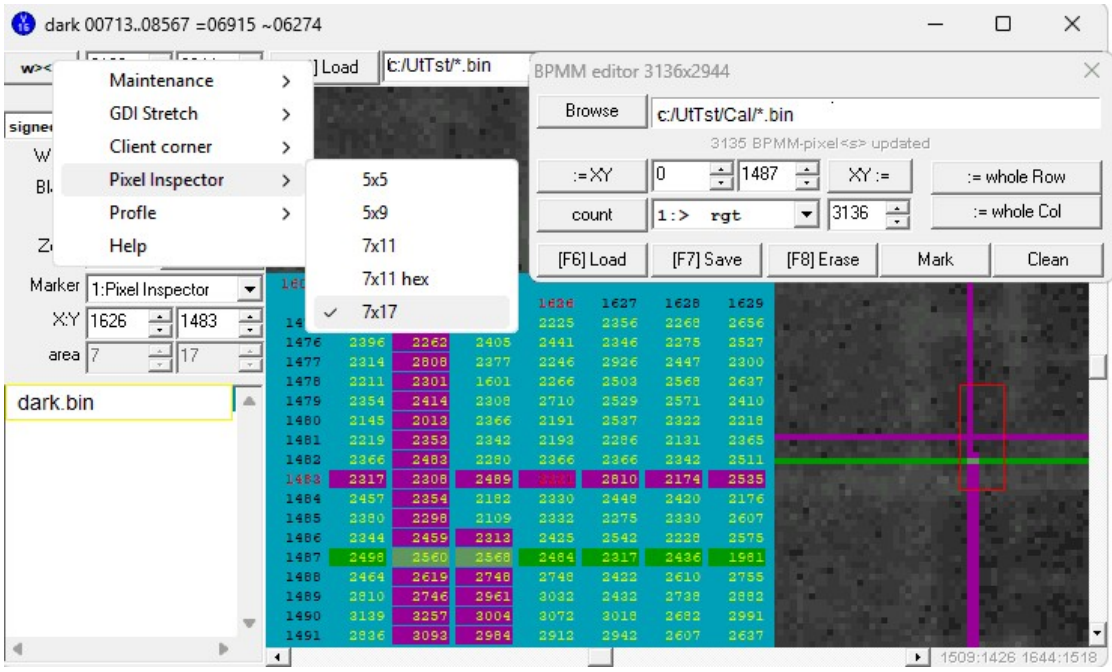
= 4:OS-specific 'open'
  - Invokes ~ShellExecute('open',image_file).

= 5,6:Hide <un>selected file_name<s> in the files'list
  This is to compare histograms, profiles, etc of two image_files, when
those aren't neighbors in files'list initially.
  - a) <by this operation> make them neighbors in the list,
  - b) <by arrow_keys up|down> move files'list_caret and watch quantitative
changes between two charts or between two image_files.") ); }
```

5. Dialog [BPMM-editor]

Infrared cameras, flat-panel X-ray -detectors, industrial optical cameras shall equalize individual $P_{ixelhoton}T_{ransfer}C_{urve}$s. The PTC of aliased pixel can't be properly equalized (by it's currrent value and pre-captured calibration-data alone). Then display of aliased pixel need a 2D-filtration (involving good neighbors). Pixel's alias could be:
= transient, where the fact of alias is detected (and then fixed) run-time;
= permanent, where the fact of alias is:
- detected automatically (from calibration data-series) in advance;
- marked manually in advance.

Respectivelly, calibration-data of those cameras <might> incapsulate two overlay-files:
- automatic BadPixelsMap,
- manual BadPixelsMap.



This dialog shows overlay **BadPixels' map** <automatic>. Shows & edits overlay **BadPixels' map manual**.

At open this dialog:
= <by current loaded image> assigns overlay's dimension,
= <tries to> loads the overlays from SDK-specific location.
= displays status line :
 ~ e.g. "BMP NNNNN pixels, BPMM doesn't exist",
 ~ e.g. "BMP doesn't exist, BPMM MMMMN pixels".

```
Editing area (marked by green color in UI-topology):
- BPMM_org    two spin_edits (x & y) specify 2D-point;
- BPMM_cnt    combo_box & spin_edit specify direction & count.

The actions:
[Browse]             selects location of loading overlays;
[:= XY]              BPMM_org := Marker_XY;
[XY :=]              Marker_XY := BPMM_org;
[count :=]           assigns BPMM_cnt by Marker_XY;
[:= whole Row]       assigns BPMM_org, BPMM_cnt by Marker_XY;
[:= whole Col]       -"-;
[Load]               loads the overlays (abandons changes);
[Save]               saves BPMM in file;
[Erase]              fills BPMM by 0 ([Erase]+[Save] delete BPMM.raw);
[Mark]               fills editing area by 1;
[Clean]              fills editing area by 0.
```

Pixels' colors and text-background:

```
- dark magenta   color -    <>0 in BPM,
- light magenta  color -    <>0 in BPMM,
- green          color -    within "editing area".
```

To enhance visibility of ~pixel-size-details on specific monitor:
 - utilize GDI-stretch selection ~= "*White|Black details*" in the pop-up menu (perhaps prefer GDI-build against GPU-build);
 - adjust "base bk-color" in [Settings].

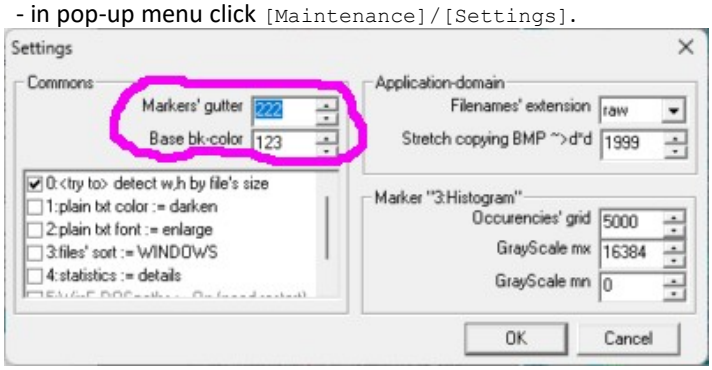
6. An example of "grayscale & profile" to comprehend a dataseries of image-processing-steps

Assume a data-series that contains either images and transient image-processing-frames (~derivations, laplacians). A debug-dumper traced:
- "0Inp" - input image
- "6Ca_Ave" - a "scrape-laplacian" of ~small magnitude
- "6Ca_Lap" - a "scrape-laplacian" of ~large magnitude

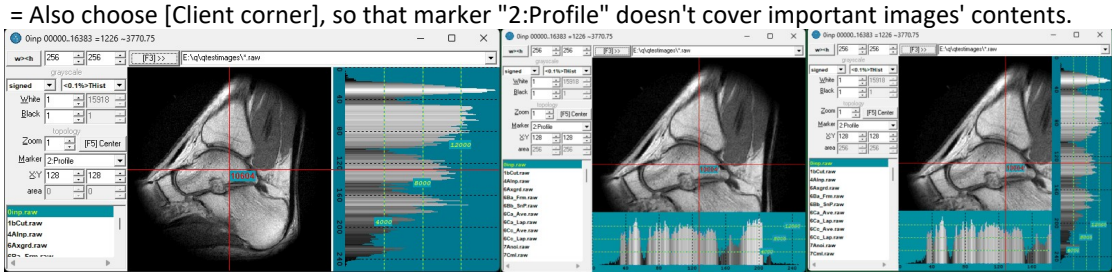
The "0Inp" is <typically> unsigned (used ~<=15 bits) and then loads the same as [unsigned] or [signed]. The "scrape-laplacians" are signed.

6.1 Initial GUI-settings

= Adjust [Markers' gutter], so that marker "2:Profile" has convenient sze:
- [F9] or right_mouse_button;
- in pop-up menu click `[Maintenance]/[Settings]`.

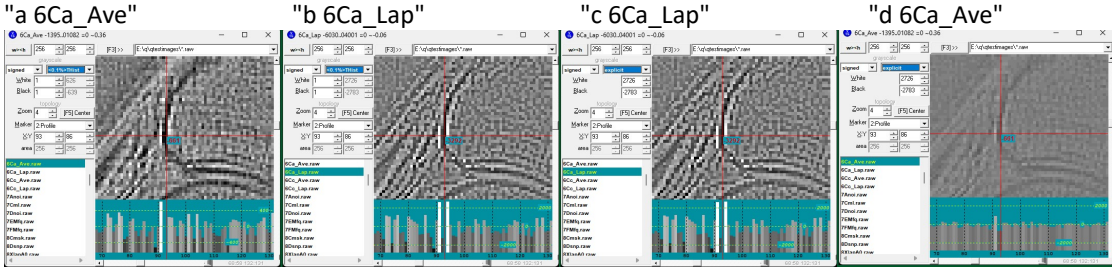


= Also choose [Client corner], so that marker "2:Profile" doesn't cover important images' contents.



6.2 Visually comprehend two laplacians of different magitude

"a 6Ca_Ave" "b 6Ca_Lap" "c 6Ca_Lap" "d 6Ca_Ave"



= compare morhology:
- select grayscale "**<0.1%> THist**" (truncate histogram in 0.1%);
- comprehend image-presentations "a", "b".

= compare magnitude:
- select laplacian of bigger magnitude ("c");
- select grayscale "**Explicit**" ([White},[Black] hide percentage-spin-edits and enable spin-edits of explicit values; the image-presentation itself remains unchanged);
- view laplacian of smaller magnitude ("d") at the same explict [White], [Black].